Genre Classification and clustering using Machine Learning

Project Report: Machine Learning for Media Technology
DM1590

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Abstract

In this project, we aim to try supervised and unsupervised machine learning models to classify music genres based on audio features extracted from the Essentia music extractor, using the AcousticBrainz music dataset. We apply dimensionality reduction techniques such as PCA, t-SNE, and UMAP to visualize the data and analyze its structure. Supervised classification models include Random Forest, Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA) are models used and compared based on accuracy and confusion matrices. Our results show that Random Forest achieves the highest accuracy, while dimensionality reduction provides insights into genre clustering and the relevance of different musical features from data. Conclusions include that models are best at finding differences between the cluster of Jazz and Classical music from the rest of the genres as well as energy-based and change-of-harmonic-based features were most important in separating the data into genres.

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1. Introduction

In this project, we explore the use of machine learning methods to analyze and classify a large music dataset. The goal is to investigate how different dimensionality reduction techniques and classification algorithms perform on genre prediction, and how unsupervised methods can help reveal the structure of the data. Due to the high number of unique genres and features, we focus on a subset of the most common genres to ensure meaningful results.

For this project the goal was to use the supervised and unsupervised machine learning models we were taught in the course to create a model that could predict genres. Since both members of the group had a common interest in music, that became an early theme that was meant to be incorporated. After searching for datasets we found the Acoustic Brainz () dataset containing musical data extracted via the Essentia music extractor from around 2 million song audio files. With the dataset at hand, we decided that the aim of the project was to create a machine learning model that could label the genre of a song given that the song actually is labeled with a genre in a set range of genres.

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2. Dataset and Preprocessing

The dataset consists roughly 2 million songs with a wide range of audio and metadata features, the dataset was gathered from AcousticBrainz[1]. To prepare the data, we selected six of the most frequently occurring genres to reduce complexity and increase performance. Numerical mappings were applied to categorical variables, and all features were scaled to the same scale to ensure fair comparison between methods. We also removed missing or complex data such as multi dimensional features since there were other features that were calculated from the complex ones.

Each song in the dataset was stored in a .JSON file containing a python dictionary where the keys were the feature names and the values to the corresponding key were either a float, string or array. The features were categorized in three levels: low-level, rhythmic and harmonic.

In order to limit the amount of labels for the supervised learning we wanted to reduce the amount of genres in the dataset to a single digit number. To solve this, we wrote a python program that read song files in order from the folder they were stored in, extracted the value for genre, if there was one, and stored the values until we reached 100 000 instances in the list. We then counted the most commonly occurring genres in that list and stored them for later use.

After this a program that read each song file in the folder they were stored in and first checked if the file had a value for the key “genre” to have a label for the supervised learning later. Following this, the genre was compared to our list of the top six most common genres. If the genre was one of the six selected genres, features were extracted and written in a new text file for the complete processed dataset, with all data linked to one song on a single line with values separated by commas. Since we only downloaded parts of the complete dataset we managed to find 60 000 songs that were in the six most common genres.

Finally we read this file containing the selected data that had been written and created a pandas dataframe from it, removing a column that had missing values for all songs and removing lines that had missing values (resulted in 5 songs out of 60 000 being removed from the total data). Each of these songs then had 27 features containing mostly mean values of different musical aspects.

We chose to include only features that were easier to handle, which means we excluded some vector time-domain data such as long sequences or spectra. This is because they either require more advanced processing or would increase the dimensionality more than wanted. For our purpose to compare different classification models and test dimensionality reduction it was more practical to use numerical features and mean values that already summarized or represented important musical aspects. All of the more complex features that were removed had mean values that we used or other features calculated from them that we used.

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[1] *AcousticBrainz Dataset*, https://acousticbrainz.org/

3. Methods

3.1 Data handling

3.1.1 Mapping and scaling the data.

The mapping was only necessary for two features; key center and mode. Since the dataset contained the 12 key centers that are found in the chromatic scale we mapped each key center to an integer from 0 to 11. Additionally we mapped the minor and major modes to 0 and 1 respectively.

After preprocessing and mapping the data it now only consisted of numerical values meaning that they only had to be scaled. This was done with SciKit Learn StandardScaler[2] which does the following calculation:

$$z = \frac{x - \mu}{\sigma}$$

subtracting the mean from the data points and dividing with the standard deviation. This gives every feature on the same scale.

3.1.2 Splitting the data

For the linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA) we split the data to 70% for training and 30% for testing.

For the random forest classifier (RF) a validation set was added to be able to find the amount of decision trees that gave the highest accuracy score. It was decided for the split to be 60% training, 20% validation and 20% for testing. The decisions to split the data in these proportions came from general guidelines in Machine Learning and data splits that were learned through the course.

3.2 Unsupervised learning

3.2.1 Principal Component Analysis (PCA)

To make the dataset easier to visualize and to see if some algorithms would perform better, we applied dimensionality reduction methods. To visualize the data and separability of genres we used PCA to reduce the data to 2D space. This was done by calculating the covariance matrix of the data and finding eigenvalues and eigenvectors from this matrix. We

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[2] *StandardScaler* https://scikit--learn-org.translate.goog/stable/modules/generated/sklearn.preprocessing.StandardScaler.html?_x_tr_sl=en&_x_tr_tl=sv&_x_tr_hl=sv&_x_tr_pto=sc

then decide the two principal components by getting the largest Eigenvalues from the data features and taking the two eigenvectors corresponding to the largest eigenvalues. The data points are then projected onto our eigenvectors to get the data in our two principal directions.

For reducing dimensions before doing supervised learning we wanted to keep most of the variance in the data, which was chosen to be 95% left after reduction. The explained variance of each principal component was calculated by dividing each eigenvalue by the sum of all eigenvalues. The number of components needed to keep 95% of the explained variance was found by adding the explained variance of each eigenvalue in descending order until the sum was greater than 95%. When the number of components needed was found, these principal components were extracted and the data was projected onto these components.

3.2.2 K-means Clustering

We used K-means clustering on the data to identify potential groupings and validate if genres form distinct clusters. The optimal number of clusters, n, was estimated by finding the number n that gave the highest silhouette score. The silhouette score was calculated with the Scikit-learn metrics module[3]. We plotted the clusters with the highest silhouette score in the first two principal directions to evaluate if the clusters represented specific genres.

3.2.3 t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a method used to visualize high-dimensional data in lower dimensions, in our case in 2D. It works by measuring how similar data points are to each other and then tries to place them in the lower dimensional space so that those similarities are preserved as much as possible[4]. It is a method based on t-distribution statistics. For a mathematical grounding see source. Since the process involves minimizing a complex cost function that isn't guaranteed to find the same solution every time, the final result can vary depending on the initial conditions.

3.2.4 UMAP Uniform Manifold Approximation and Projection (UMAP)

UMAP is a method for reducing the dimensions of data. It is based on the assumption that the data lies on an underlying structure, or "manifold", and attempts to preserve both local and global patterns when reducing the data to fewer dimensions[5]. Unlike t-SNE, which often focuses on local structures, UMAP has the advantage of preserving more of the overall

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[3] *Scikit-learn: Silhouette score, https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_samples.html*
[4] *Scikit-learn: t-SNE*, https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html
[5] *UMAP Documentation – How UMAP works*, https://umap-learn.readthedocs.io/en/latest/how_umap_works.html

structure of the data. The result is a low-dimensional representation that is often well suited for visualization and clustering.

3.3 Supervised Learning

3.3.1 Random Forest Classifier (RF)

Random forest classifier is an estimator that fits a number of decision trees on subsets of data and averages their scores to improve accuracy and reduce overfitting. We used the Scikit-learn model of this classifier[6] to label genres on our data.

3.3.2 Quadratic Discriminant Analysis (QDA)

QDA is a classification method that models each class with its own shape (covariance). This makes it flexible and good when your data classes have different spreads or patterns.

3.3.3 Linear Discriminant Analysis (LDA)

LDA is a simpler method than QDA. It assumes that all classes share the same spread and tries to find the best linear combinations of features that separate them. It works well when the data fits these assumptions. We used Scikit-learn for both the LDA and QDA models[7].

3.3.4 Evaluation

For the evaluation of our different supervised learning algorithms the decision was made to compare according to accuracy score as well as confusion matrices. The matrices were normalized by dividing each value in a row by the sum of that row to give a percentage scale.

4. Results and Evaluation

Clustering with K-means on our data showed that the best silhouette score was achieved for K = 2.

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[6] *Scikit-learn: Random Forest, https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html*

[7] *Scikit-learn: LDA and QDA*, https://scikit-learn.org/stable/modules/lda_qda.html

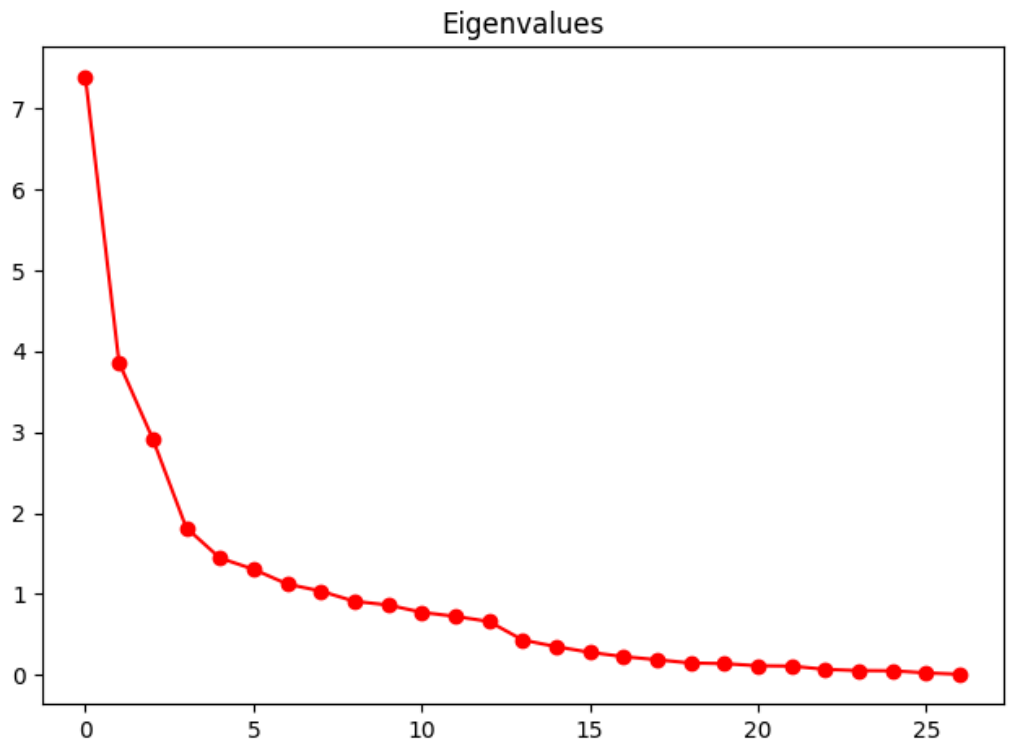

Figure 1: Sorted eigenvalues of covariance matrix.

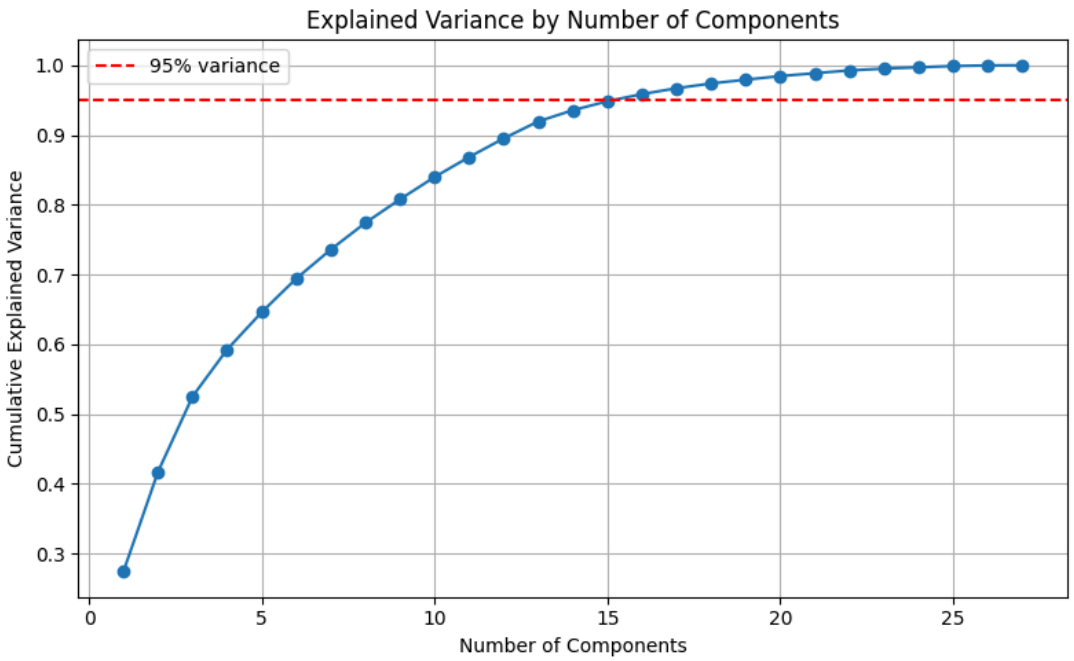

Figure 2: Explained Variance in data by number of principal components.

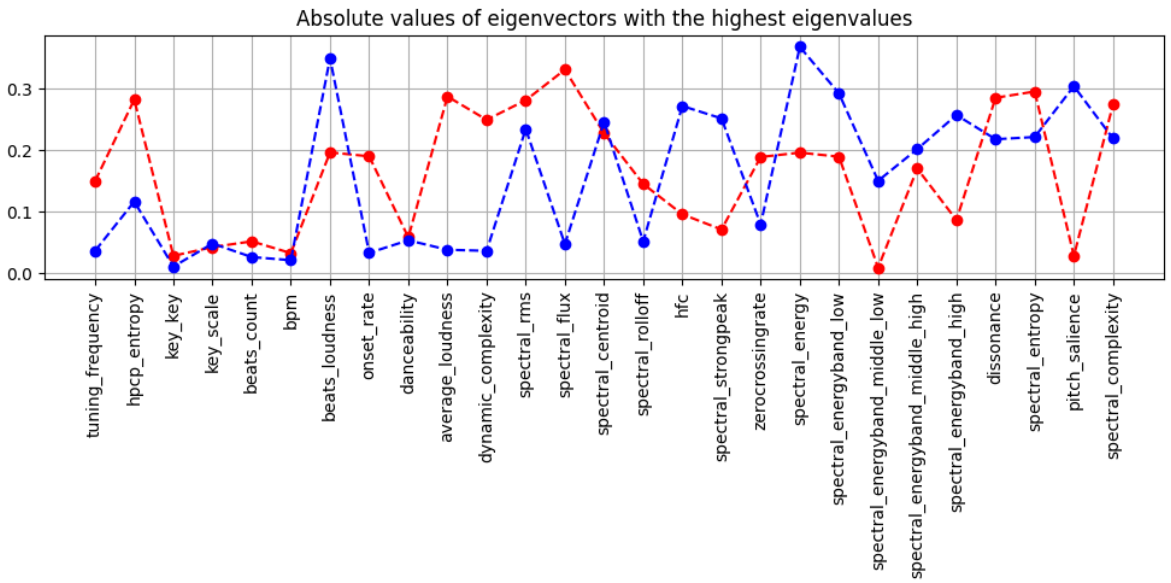

Figure 3: Absolute values of elements in first and second principal components. The red line is PC1 and the blue line is PC2.

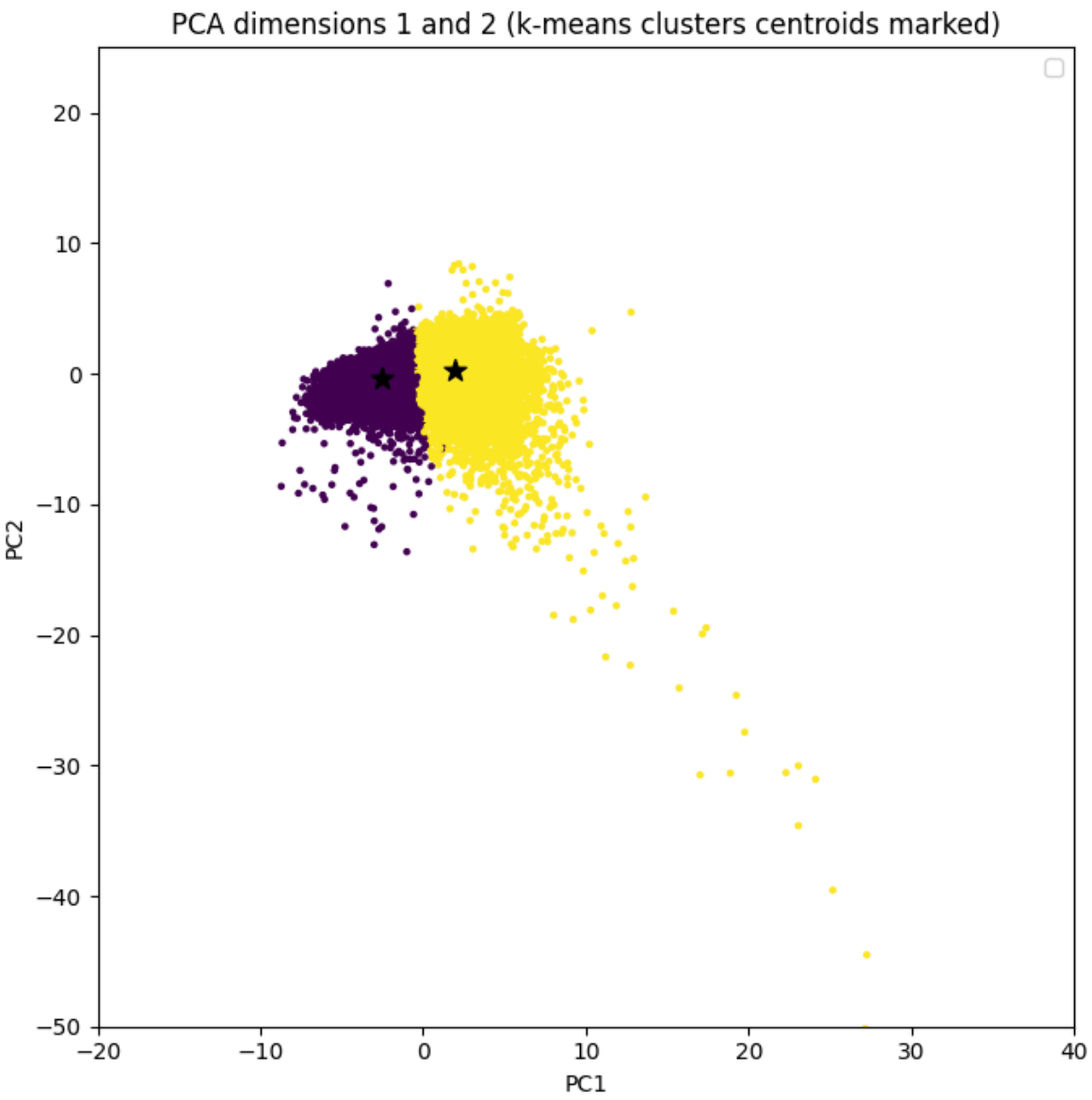

Figure 4: K-means clusters and centroids projected in first and second principal components.

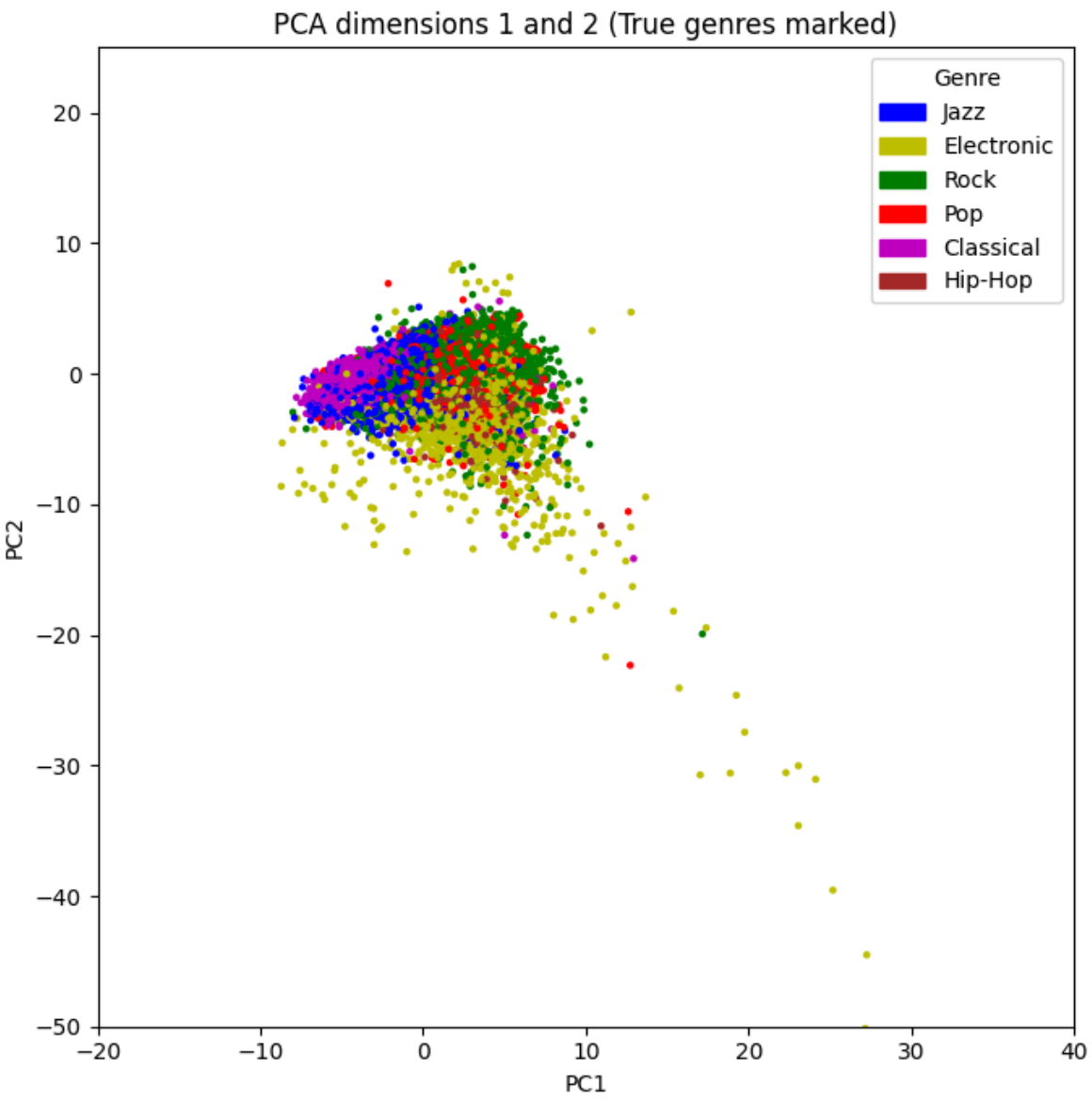

Figure 5: Data projected in first and second principal components with true genres marked.

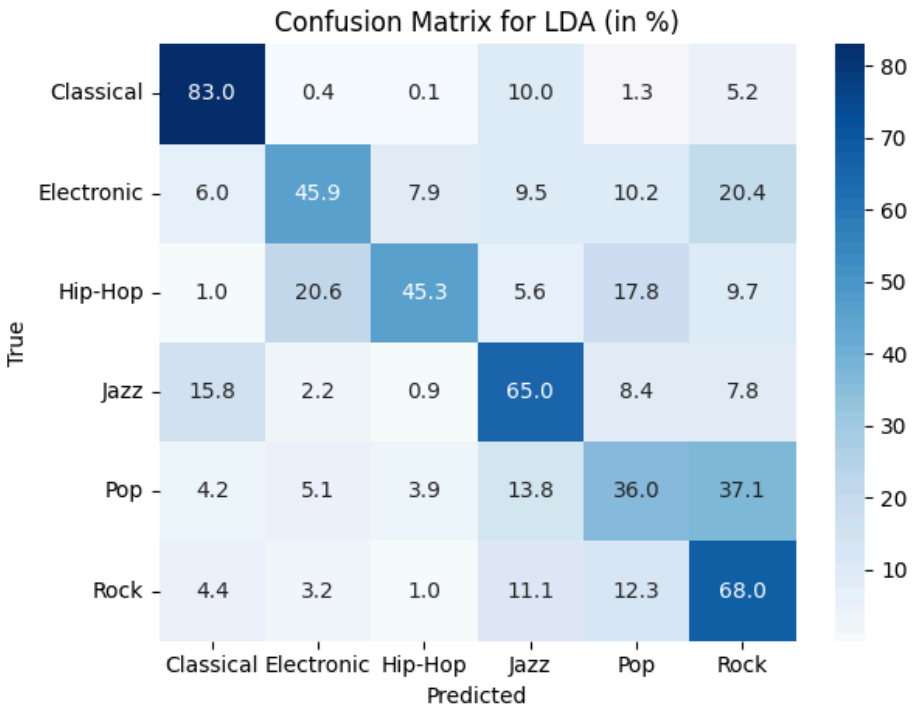

Figure 6: Confusion matrix in % for LDA.

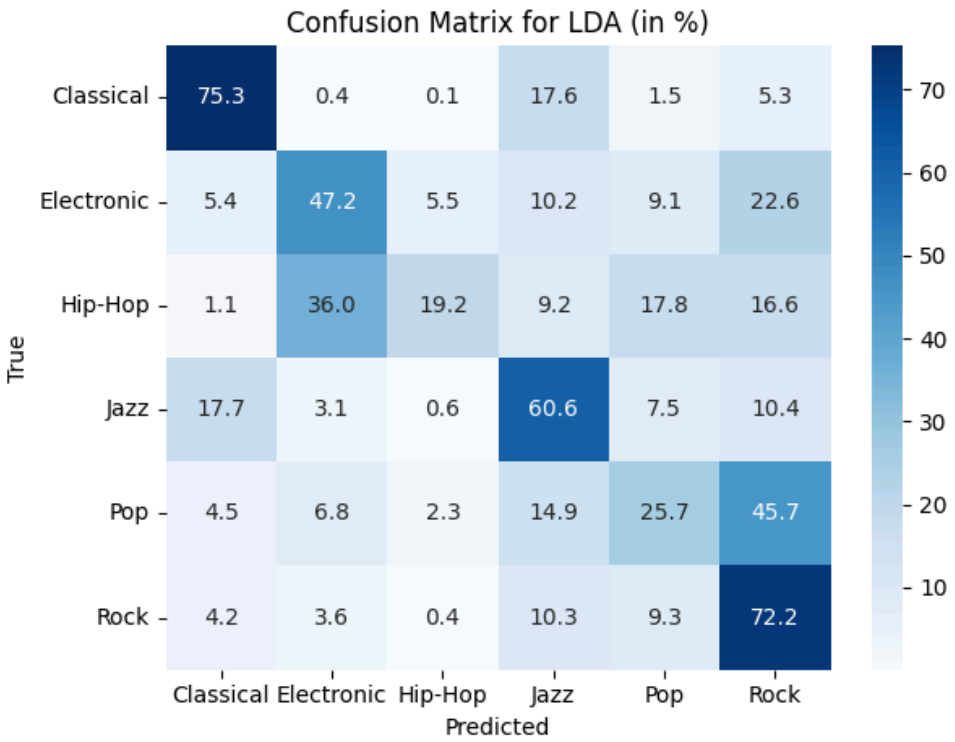

Figure 7: Confusion matrix in % for LDA after PCA (15 components).

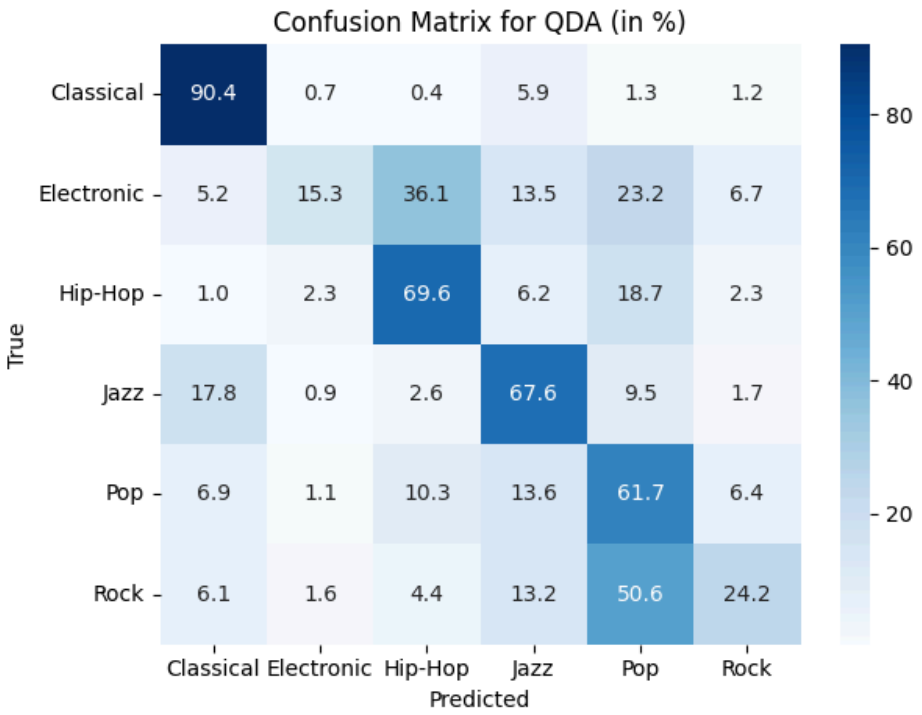

Figure 8: Confusion matrix in % for QDA.

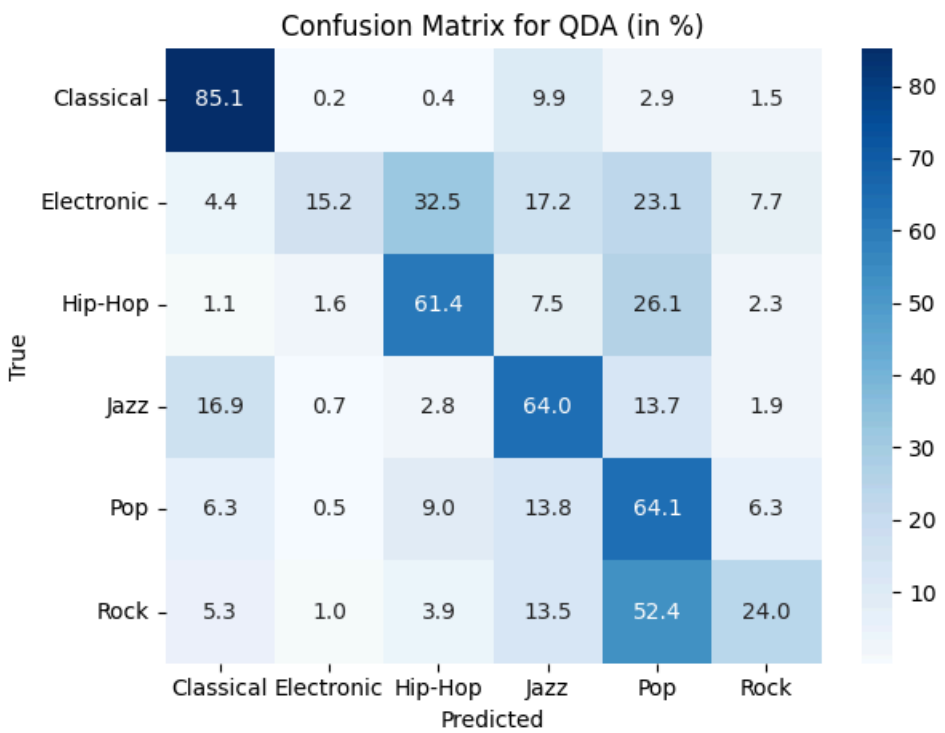

Figure 9: Confusion matrix in % for QDA after PCA (15 components).

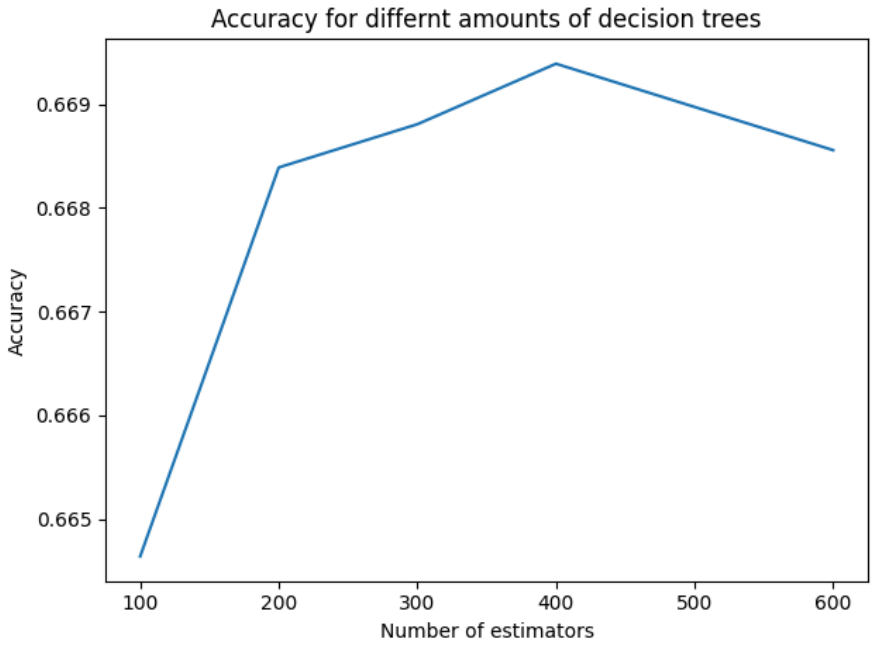

Figure 10: Accuracy of RF on validation set for different number of decision trees.

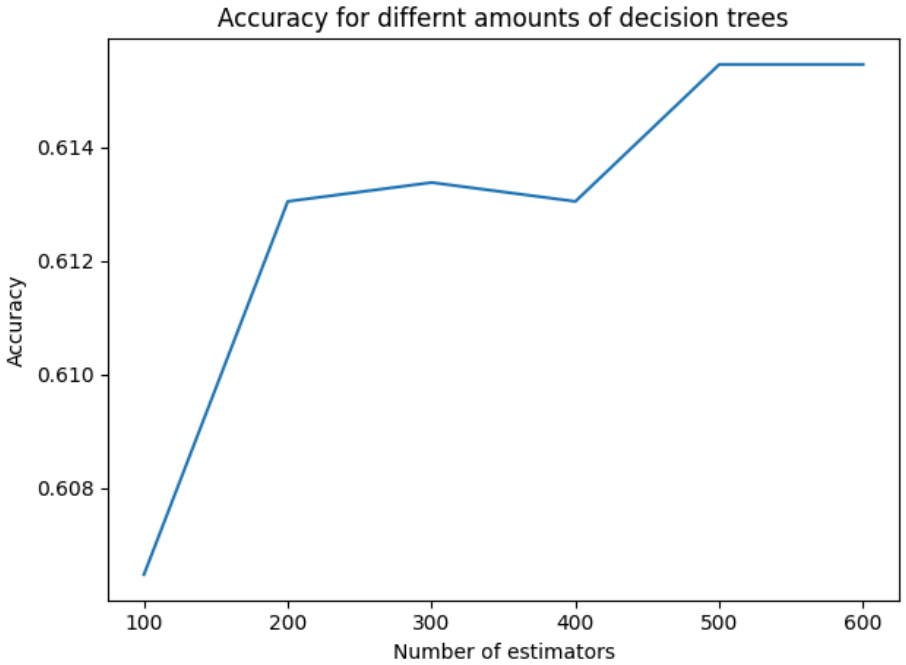

Figure 11: Accuracy of RF after PCA (15 components) on validation set for different number of decision trees.

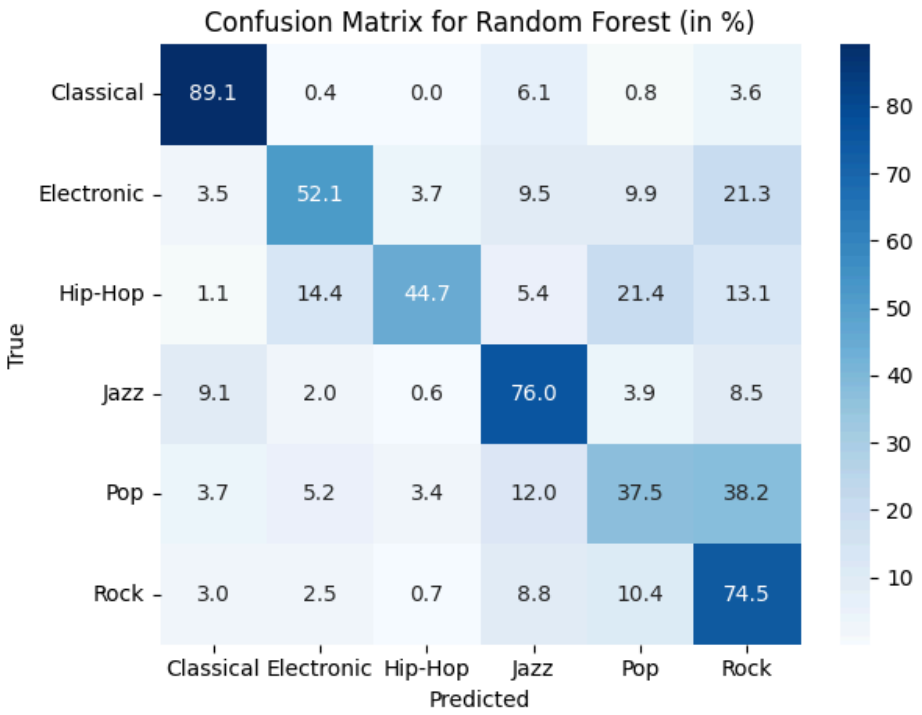

Figure 12: Confusion matrix in % of RF.

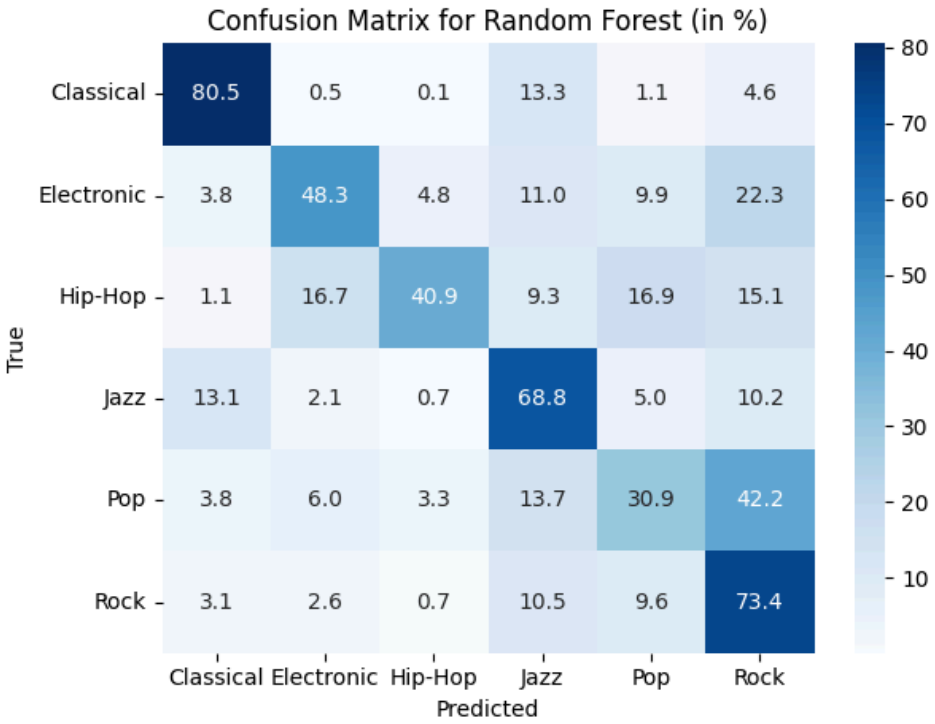

Figure 13: Confusion matrix in % for RF after PCA (15 components).

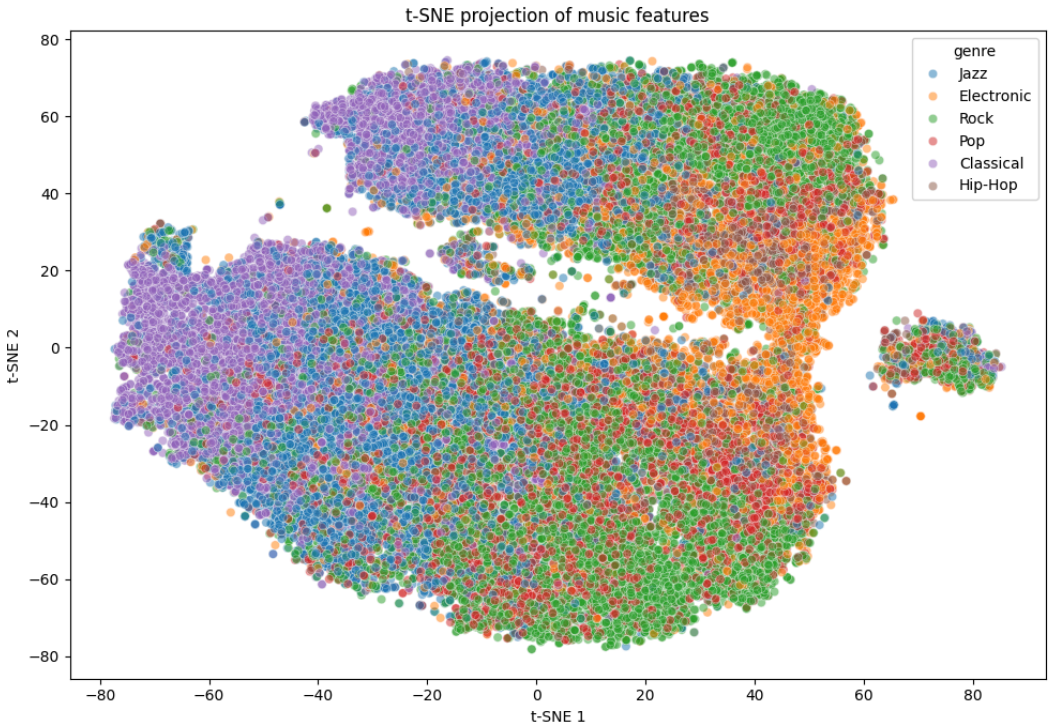

Figure 14: t-SNE Projection in two dimensions.

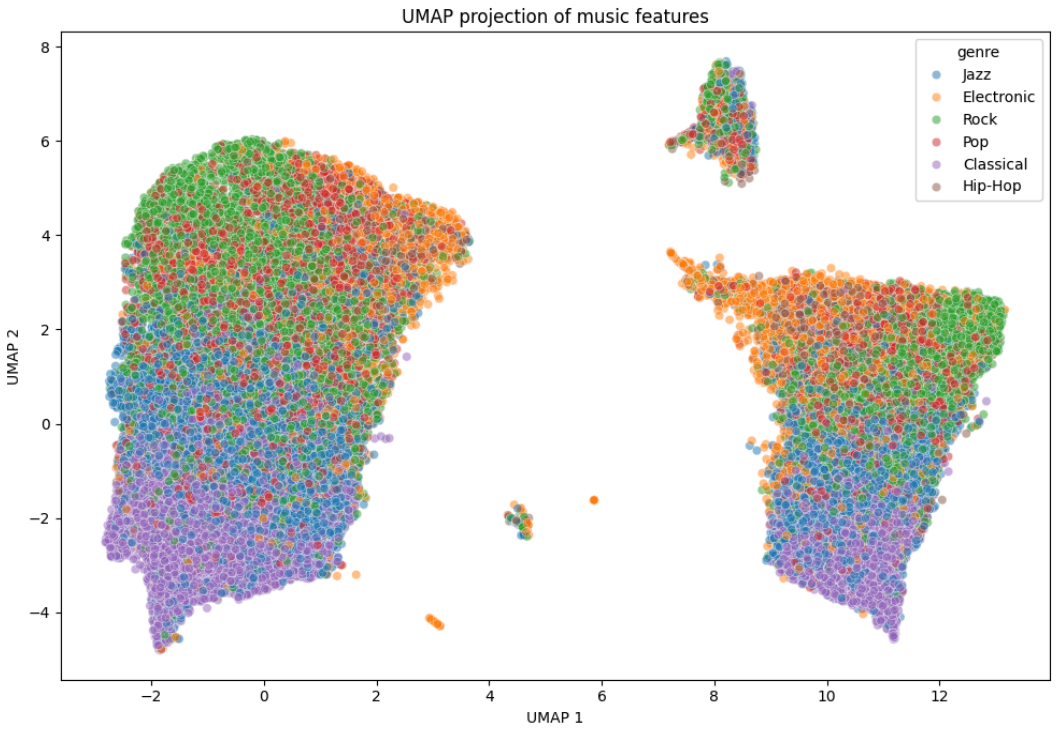

Figure 15: UMAP Projection in two dimension

Model:	**Accuracy Score:**
LDA	0.6119
LDA + PCA	0.5714
QDA	0.5342
QDA + PCA	0.5164
RF (400 decision trees)	0.6735
RF + PCA (600 decision trees)	0.6222

Table 1: Accuracy score for our different models.

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5. Discussion

When evaluating the results from our K-means clustering, the value of K (K=2) clusters that gave the highest silhouette score might already be telling in how well the method worked. The best value k was for two clusters and since we know from our dataset that the data contained songs from six genres, this indicates that our model is under fitted. Comparing figure 4 and 5 gives the information that in the first two principal directions, the data in the left cluster of figure 4 seem to consist of Jazz and Classical music while the rest of the genres are in the right cluster of figure 4. Therefore, the K-means clustering seems to be able to create clusters of Jazz or Classical songs and non-Jazz or non-Classical songs.

Random Forest was our best-performing model when we ran it without dimensionality reduction. We tested different numbers of trees and found that 400 decision trees gave the best results, which we chose as our final parameter. The model works by building many different decision trees and then making majority decisions from them, which makes it robust against overfitting and often very stable in its predictions. Compared to the other models, it gave by far the highest accuracy, and was especially good at classifying classical music, jazz and rock correctly.

The worst performing model was QDA after PCA reduction. This makes sense if we evaluate results and how the methods work since we lost structure and information when we projected the data. This loss of data hurt performance in our case, performance dropped around 2% but QDA already performed the worst out of our supervised models. The relative performance between genres seems to be consistent between QDA with PCA data and our complete dataset.

The LDA model were in the middle performance wise amongst our supervised models. Without PCA the accuracy was around 61% and with PCA it dropped around 4% points. Compared to QDA however, the relative performance between genres dropped significantly after PCA. This is not really seen in the other supervised models. For example the accuracy for Hip-hop goes from 45.3% to 19.2% after using PCA data. One reason why this could happen is because PCA has removed dimensions that are low in variance but still might be highly discriminative between classes. So we seem to have removed information required by our LDA to effectively distinguish Hip-Hop as a genre.

Evaluating all of the confusion matrices combined with the conclusion of the K-means clustering being good at finding Jazz or Classical songs we can see that for all songs with the true genre being Classical, the incorrectly guessed genre that is most common is Jazz and vice versa. Therefore, the data seems to differ between songs that are Jazz or Classical compared to the rest while Jazz and Classical music. Therefore the models seem to be good at classifying if a song is Jazz or Classical or if it is another genre but is not as good at classifying specific genres.

Pop was the genre that all models, except QDA, had the most difficulty classifying correctly. We saw this clearly in our confusion matrices where pop was the most confused with other genres. This may be because pop is a fairly broad genre spectrum and often shares characteristics with other types of music. This makes it difficult for the model to find clear, unique patterns that only belong to pop. It is also possible that the music that is tagged as pop in the dataset is more mixed than other genres, which means that the model does not get a clear pattern to learn from.

When visually analysing the t-SNE and UMAP with the previous statement, it makes sense that Pop is the most confused genre. In both figure 14 and 15, we can see that songs that have Pop as their genre, marked in red, seem to be scattered the most in the plot compared to the other colored genres.

Further analysing the t-SNE and UMAP projections, there are not really any clear clusters that group the genres together in the two dimensional projections. In both projections there seems to be two large clusters and one small cluster, in the bigger clusters the genres are somewhat separated on the x-axis in figure 14 and y-axis in figure 15. This means that one of the projected directions could be useful for classifying genre although not perfect since for example pop is quite scattered all over the clusters. The small clusters do not really seem to visually help classify genres since they contain data from all genres.

Our most prominent features could be observed from our two principal directions after visualizing our data (in other words projecting our data down to 2D). The most prominent features from our two principal directions was **Spectral Energy** and **Beats_Loudness** for one principal direction. For the other principal direction it was **Spectral_Entropy** and **Spectral_Flux.**

This result suggests that genre separation in the dataset is highly dependent on energy and loudness of the songs. It is linked to how loud or intense the Essentia music extractor perceives the song to be. For the other principal direction the flux and entropy relates more to harmonic information, how unpredictable the song is or what the rate of change in the song is. This can really help differentiate between more stable structures (Classical and

Jazz) compared to songs with more noisy or dynamic aspects (Rock). This insight is interesting because the statistical and Machine Learning methods align with what one would expect from a purely musical perspective on separating between genres.

Before summarizing our conclusions, one drawback that could have improved our results if implemented are the time-domain data. If implemented, the models could have found if features related to song structure or change of features throughout a song contribute to better accuracy of classification.

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6. Conclusion

The conclusions drawn from this project are that our unsupervised machine learning models seem to only differentiate our data into two clusters containing Jazz as well as Classical music in one and the other genres in the other. The supervised models seem to be able to classify specific genres at best with 67% accuracy score with Classical, Jazz and Rock music being easiest to predict. The main components of the musical data that contributed to separating genres seem to be energy of a song and the rate of change in harmonic information.

This project shows that supervised learning methods, particularly Random Forest, can classify music genres with reasonable accuracy when applied to well-preprocessed data. Dimensionality reduction techniques are valuable for understanding the structure of music data but not as useful for classification. The inclusion of time-domain vectors in our features would also be an interesting addition for future work to see if that data contains important information about the different genres.

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7. References

1. *AcousticBrainz Dataset*, https://acousticbrainz.org/ [Accessed 2 May 2025].

2. *StandardScaler* https://scikit--learn-org.translate.goog/stable/modules/generated/sklearn.preprocessing.StandardScaler.html?_x_tr_sl=en&_x_tr_tl=sv&_x_tr_hl=sv&_x_tr_pto=se

3. *Silhouette score,* https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_samples.html [Accessed 10 May]

4. *Scikit-learn: t-SNE*, https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html [Accessed 12 May 2025].

5. *UMAP Documentation – How UMAP works*, https://umap-learn.readthedocs.io/en/latest/how_umap_works.html [Accessed 12 May 2025].

6. *Scikit-learn: Random Forest,* https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html [Accessed 12 May]

7. *Scikit-learn: LDA and QDA*, https://scikit-learn.org/stable/modules/lda_qda.html [Accessed 10 May 2025].